

Putnam 1985 — A2

Jonathan Kasongo

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Let T be an acute triangle. Inscribe a rectangle R in T with one side along a side of T . Then inscribe a rectangle S in the triangle formed by the side of R opposite the side on the boundary of T , and the other two sides of T , with one side along the side of R .

For any polygon X , let $A(X)$ denote the area of X . Find the maximum value, or show that no maximum exists, of

$$\frac{A(R) + A(S)}{A(T)},$$

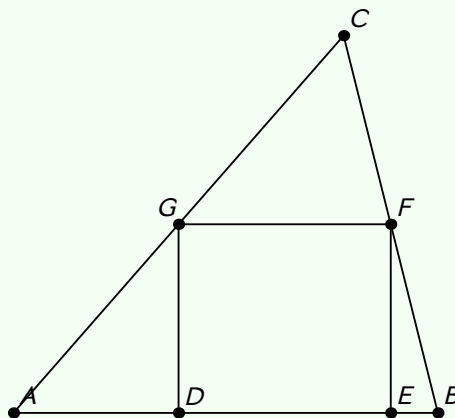
where T ranges over all triangles and R, S over all rectangles as above.

Solution

The answer is $\frac{2}{3}$. We begin with this well-known fact.

Claim 1: Let T_0 be an acute, non-degenerate triangle. Inscribe a rectangle R_0 with one of its sides lying on a side of T_0 . Then $A(R_0) \leq \frac{1}{2}A(T_0)$.

Proof: Orient the diagram such that the side of T_0 that R_0 lies on is the base. Say the base of T_0 has length b , and T_0 has height h . There is a triangle T'_0 formed by the region bounded by the two sides that R_0 doesn't lie on and the line passing through the side of R_0 opposite to the one that lies on the base of T_0 .



For example here $T'_0 = \triangle CGF$. Note that $\angle CAB = \angle CGF$ and $\angle CBA = \angle CFG$ so $T'_0 \sim T_0$ by the AAA criterion. So let the base of T'_0 be λb and the height be λh for $(0 < \lambda < 1)$. The area of R_0 is

$$(\lambda b)(h - \lambda h) = bh \cdot \lambda(1 - \lambda).$$

Now it can easily be checked that $f(\lambda) = \lambda(1 - \lambda)$ has maxima $f(1/2) = 1/4$. Thus $A(R_0) \leq \frac{1}{4}bh = \frac{1}{2}A(T_0)$.

□

Take the same setup as claim 1. The area of T'_0 is $\frac{1}{2}(\lambda b)(\lambda h) = \frac{1}{2}\lambda^2 bh$. If we inscribe a rectangle R'_0 inside of T'_0 then it has a maximum area of $\frac{1}{4}\lambda^2 bh$. due to claim 1.

So for some selection of λ the maximal value of $A(R_0) + A(R'_0)$ is

$$\begin{aligned} &= bh \cdot \lambda(1 - \lambda) + \frac{1}{4}\lambda^2 bh \\ &= bh \left[-\frac{3}{4}\lambda^2 + \lambda \right] \end{aligned}$$

We can easily check that $g(\lambda) = -\frac{3}{4}\lambda^2 + \lambda$ has maxima $g(2/3) = 1/3$. Thus

$$\max \frac{A(R_0) + A(R'_0)}{A(T_0)} = \frac{bh/3}{bh/2} = \frac{2}{3}. \blacksquare$$